

Project RANG: Comprehensive Execution & Phase Report

1. Problem Statement

Project Rang was initiated to address a massive, interconnected environmental and socioeconomic crisis.

- **Step 1: Identifying the Environmental Crisis.** India generates roughly 30 million tons of banana agricultural waste annually. Most of this pseudostem biomass is either burnt or discarded, while simultaneously, the traditional cotton textile industry depletes severe amounts of water.
- **Step 2: Addressing Artisan Marginalization.** Rural women possess exceptional, latent traditional skills, yet they operate in a market where they lack direct access to high-end, global consumers. This keeps them dependent on exploitative middlemen.
- **Step 3: Recognizing the Consumer Gap.** Conscious consumers actively want to purchase ethically made goods but are currently stuck between aggressively overpriced luxury eco-brands and highly pollutive, low-quality mass production.

2. Market Research

The project began as an Enactus internal ideathon concept for an Indian craft marketplace, which was entirely pivoted following our Phase 1 research into agro-waste.

- **Step 1: Material Feasibility Analysis.** We conducted deep technical research to determine if banana fabric could withstand the daily wear-and-tear typically handled by cotton. The research confirmed that banana fiber is highly breathable, UV-resistant, and crucially, up to 3x stronger than standard cotton.
- **Step 2: Supply Chain Logistics Mapping.** We mapped out a localized supply chain, identifying regional banana wholesalers and small decortication machine operators in rural clusters capable of extracting the fiber.
- **Step 3: Sourcing Partnerships.** We researched integration with established NGO networks, establishing a pathway to source post-harvest pseudostems directly from farmers.

3. Customer Insights

We deployed primary survey instruments to validate our market approach and establish a data-backed pricing strategy.

- **Step 1: Purchase Intent Validation.** Survey data revealed high market readiness, with 77% of respondents stating they were "Likely" or "Very likely" to purchase banana fiber products regularly.
- **Step 2: Analyzing the "Impact" Hook.** We found that 61.5% of consumers are strongly and positively influenced to purchase when they know a product actively utilizes agricultural waste and prevents field-burning.
- **Step 3: Formulating the Pricing & Marketing Strategy.** Price elasticity data showed 46.2% of consumers expect price parity with conventional goods, while another 46.2% are willing to pay a 10-20% premium. The strategic conclusion was to lead all marketing with the "3x Stronger" technical performance claim to convert users, using the "Zero-Waste" story to build brand loyalty.

4. Business Model

The project was structured as an asset-light, Direct-to-Consumer (D2C) social enterprise centered around a circular ecosystem.

- **Step 1: Designing the Zero-Waste Loop.** We mapped the utility of the entire pseudostem. The extracted fiber (1.5-2%) is utilized for high-value textiles, while the remaining waste material (85-90%) is processed into fabric pulp.
- **Step 2: Establishing D2C Direct Market Linkage.** By bypassing traditional retail middlemen, the model ensures that 60-70% of the profit margin flows directly back to the women artisans. This model increases the artisans' average monthly income by 200%.
- **Step 3: Calculating Unit Economics.** We established strict profitability targets. A standard handcrafted tote bag costs ₹340 to produce and sells for ₹540, generating a 35.4% profit margin. A standard double bedsheet costs ₹1690 to produce and sells for ₹2500, generating a 32.4% margin.

5. Product Development

Following logistical validation, we translated our research into physical product lines.

- **Step 1: The Textile Line.** We engineered our first commercial line using a specific banana-cotton blend designed for comfort and durability, featuring upcycled tote bags, scrunchies, and hand-painted bedsheets.
- **Step 2: The Packaging Innovation.** Heavy R&D was deployed to solve the issue of the leftover banana pulp. We developed a method to mold the waste pulp into biodegradable packaging that is significantly more cost-effective than paper or plastic and boasts superior tear resistance (3x stronger than wood pulp). We designed the custom packaging molds using Blender.
- **Step 3: Customization Engineering.** To prevent overproduction and maintain exclusivity, we implemented a "paint-to-order" model, allowing for bespoke, small-batch releases.

6. Prototype Testing

The transition from theory to tangible product involved strict quality assurance and iterative design.

- **Step 1: Pilot Fabrication.** We initiated a 7-day workshop with 5-10 lead artisans, utilizing outsourced "cottonized" banana fiber from suppliers like Canvaloop to rapidly create our first 5 samples.
- **Step 2: Quality Assurance.** The initial banana fiber bedsheets and bags were stress-tested specifically for shrinkage, dye-fastness, and overall durability.
- **Step 3: Iterative Refinement.** We took this v1 prototype to multiple early-stage national competitions. While we did not win, we aggressively collected feedback from industry judges, deep-dived into similar market competitors, and refined our textile blends and supply chain efficiency to create a superior v2 prototype.

7. Website & Branding

A premium digital ecosystem was constructed to support the D2C retail strategy.

- **Step 1: Visual Identity & Storytelling.** We developed a comprehensive brand kit, complete with a professional logo, poster assets, and high-quality model photoshoots. Content was heavily optimized for Instagram, which we identified as driving 40% of our sales traffic.
- **Step 2: E-Commerce Architecture.** We built and deployed a custom e-commerce website allowing for direct buyer customization and retail.
- **Step 3: Traceability Tech Integration.** We engineered a "Digital Passport" system. Every product ships with a scannable QR code tag. Upon scanning, the buyer is taken on a digital journey showing the exact profile and story of the artisan who made their product (e.g., "My name is Sunita...") fostering deep brand trust.

8. Industry Validation

The launch of our v2 prototype triggered significant external validation and operational scaling.

- **Step 1: On-the-Ground Partnerships.** We formally secured a partnership with the established NGO Goonj to manage the collection of banana pseudostems from farmers and facilitate local artisan operations.
- **Step 2: B2B Expansion.** We expanded beyond individual retail, receiving multiple corporate offers for eco-friendly gifting sets combining our tote bags and scrunchies.
- **Step 3: Secondary Market Interest.** The development of our banana pulp packaging generated independent interest from other sustainable projects and brands looking to purchase our biodegradable packaging in a B2B format.

9. Competition Results

The continuous refinement of our business plan and prototype resulted in massive competitive success.

- **Step 1: Applying Feedback.** By overhauling our pitch deck to emphasize our unit economics, tech-enabled traceability, and packaging innovation, we created a highly defensible business plan.
- **Step 2: The Breakthrough.** Project Rang dominated the later competition circuit, ultimately winning the Abhivyakt Business Plan Grand Prize of ₹1.5 Lakhs.
- **Step 3: Continued Dominance.** Capitalizing on this momentum, the project went on to secure victories in 5 additional national competitions, proving the viability and scalability of the model.

10. Key Learnings

- **Step 1: The Power of the Pivot.** The willingness to abandon our original craft-marketplace idea in favor of a complex, agro-waste tech solution was the defining factor in the project's success.
- **Step 2: Performance Drives Sustainability.** Consumers are drawn to the eco-friendly narrative, but they purchase based on quality. Leading with the technical superiority of the material ("3x stronger") is a far more effective conversion tool than relying on sustainability alone.
- **Step 3: Innovate the Waste.** Forcing ourselves to figure out what to do with the 85% of the banana stem that isn't fiber directly resulted in our most scalable innovation: ultra-durable banana pulp packaging.